



WORK SHEET - 25

DETERMINANT & MATRICES

Q.1 (C)

$f'(x) = 0 \Rightarrow f(x)$ is constant

$\therefore |f(1)| + |f(2)| + \dots + |f(30)| = 60$. **Ans.**

Q.2 (D)

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad \dots(1)$$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \dots(2)$$

$$-\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad \dots(3)$$

$$(1) + (2) + (3) \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 3 \quad \dots(4)$$

From (1) and (4), $\frac{z^2}{c^2} = 1 \Rightarrow z = \pm c$

Hence, $x = \pm a$, $y = \pm b$, $z = \pm c$.

So, number of solution $(x, y, z) = 2 \times 2 \times 2 = 8$. **Ans.**

Aliter: Let $\frac{x^2}{a^2} = X$, $\frac{y^2}{b^2} = Y$ and $\frac{z^2}{c^2} = Z$, then the given system of equations is

$$X + Y - Z = 1, \quad X - Y + Z = 1, \quad -X + Y + Z = 1,$$

Here, the coefficient matrix

$$A = \begin{bmatrix} 1 & 1 & -1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{bmatrix} = -4 \Rightarrow |A| \neq 0.$$

$\therefore$ On solving, we get

$$X = Y = Z = 1$$

Hence, $x = \pm a$, $y = \pm b$, $z = \pm c$.

So, number of solution $(x, y, z) = 2 \times 2 \times 2 = 8$. **Ans.**



Q.3 (C)

$$\Delta_2 = \Delta_1^2$$

$$\Delta_3 = \Delta_2^2 = \Delta_1^4$$

$$\text{Hence } \Delta_1^2 = \frac{\Delta_3}{\Delta_2}$$

Q.4 (A)

$$X = P^T[(PAP^T)(PAP^T)\dots(PAP^T)]P$$

$$= A^{2005} = A^{2004} \cdot A = A \quad \text{Ans.}$$

Note : If k is the period of A $\Rightarrow A^{nk+1} = A$ for $n \in \mathbb{I}$.

Q.5 (D)

$$\text{We have } \bar{a}_{vu} = \sin(\theta_v - \theta_u) - i \cos(\theta_v - \theta_u) = -\sin(\theta_u - \theta_v) - i \cos(\theta_u - \theta_v) = -a_{uv}$$

$$\Rightarrow \bar{a}_{uv}^T = -a_{uv}$$

$$\text{Hence } M = -\bar{M}^T.$$

Q.6 (C)

$$\text{We have } S = 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2^2} + 3 \cdot \frac{1}{2^3} + \dots \infty$$

$$\frac{S}{2} = 0 + 1 \cdot \frac{1}{2^2} + 2 \cdot \frac{1}{2^3} + \dots \infty$$

Subtract

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \infty = \frac{\frac{1}{2}}{1 - \frac{1}{2}} \Rightarrow S = 2. \text{ Ans.}$$

Q.7 (C)

$$|A| = x + y + z$$

$$\det.(\text{adj.}(\text{adj. } A)) = (\det. A)^{(n-1)^2} = (\det. A)^4 = 2^8 \cdot 3^4 = 12^4$$

$$|A| = 12$$

$$\Rightarrow x + y + z = 12$$

Using beggar method $x + y + z = 9$.

$$\text{Number of matrix} = {}^{11}C_2 = 55. \text{ Ans.}$$

Q.8 (A)

$$\text{Let } A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \text{ where } a_{ij} \in \{0, 1\}$$

As, trace $A = 1$

So, any two elements in main diagonal are 0 and one element is 1. Also, non-diagonal elements can be 0 or 1.

So, number of matrices $= 3 \times 2^3 = 3 \times 8 = 24$. **Ans.**

Q.9 (B)

Given $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$\Rightarrow a + b + c = 0$ then $at^2 + bt + c = 0$ has one root $t = 1$.

Also, $D = b^2 - 4ac$
 $= (-a - c)^2 - 4ac$
 $= (a - c)^2 > 0$ (As $a > c$)

As, $a > b > c$

$\Rightarrow a > 0$ and $c < 0$

So, product of roots $= \frac{c}{a} = \frac{-ve}{+ve} < 0$

$\Rightarrow$ roots are opposite in sign.

Q.10 (D)

We must have

$$AB = I$$

$$\Rightarrow \begin{bmatrix} 5\alpha & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 10\alpha - 2 & 5\alpha \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \Rightarrow \alpha = \frac{1}{5} \text{ Ans.}$$

Q.11 (D)

Q.12 (A)

Q.13 (D)

(i) $A = \text{adj } B - B^T$ (i)

$B = \text{adj } A - A^T$ (ii)

$A^T = (\text{adj } B)^T - B$ (iii)

$\text{adj } A = \text{adj } B^T$ (from (ii) and (iii))

Since A is non-singular $\Rightarrow B$ is non-singular

$\Rightarrow A = B^T$ or $A = -B^T$.

Case-I : If $A = B^T \Rightarrow \text{adj } B = 2A \Rightarrow |B|^2 = 8|B|$

$\Rightarrow |B| = 8$ and $|A| = 8$.

Case-II : If $A = -B^T \Rightarrow |A| = -|B|$ and $\text{adj } B = 0$

$\Rightarrow |B|^2 = 0$ (Rejected)

$$(ii) \quad |\text{adj. } (2B^{-1})| = \frac{2^{3 \times 2}}{8^2} = \frac{64}{64} = 1$$

$$(iii) \quad \begin{aligned} \text{Now, } \text{adj. } B &= 2A \\ (\text{adj. } B)B &= 2AB \\ |B| I &= 2AB \\ AB &= 4I \end{aligned}$$

Q.14 (BCD)

$$P_1: x + 3y + 2z = 6$$

$$P_2: x + ay + 2z = 7$$

$$P_3: x + 3y + 2z = b$$

Note that the planes P_1 and P_3 are parallel hence system of equation will never have unique solution.

If $b = 6$ and $a = 4$ then P_1 and P_3 are identical and P_2 will intersect P_1 and P_3 at infinite number of points because they have common line of intersection.

If $b \neq 6$, then P_1 and P_3 are parallel and system will have no solution for every $a \in \mathbb{R}$.

Aliter: Given, $x + 3y + 2z = 6$ (1)

$$x + ay + 2z = 7$$
(2)

$$x + 3y + 2z = b$$
(3)

(A) If $a = 2$, then $\Delta = 0$ so unique solution is not possible. $\Rightarrow$ **(A) is incorrect.**

(B) If $a = 4$, $b = 2$; consider the equation (1) and (2), we get

$$x + 3y = 6 - 2z$$

$$x + 4y = 7 - 2z$$

$$\therefore y = 1 \text{ and } x = 3 - 2z$$

Now, on substituting in equation (3), we get

$$3 - 2z + 3 + 2z = 6, \quad (\text{True})$$

$\Rightarrow$ infinite solution exists. $\Rightarrow$ **(B) is correct.**

(C) If $a = 5$, $b = 7$; consider the equation (2) and (3), we get

$$x + 5y = 7 - 2z$$

$$x + 3y = 7 - 2z$$

$$\therefore y = 0 \text{ and } x = 7 - 2z$$

Now, on substituting in equation (1), we get

$$7 - 2z + 2z = 6 \quad (\text{False})$$

$\Rightarrow$ No solution exists. $\Rightarrow$ **(C) is correct.**

(D) If $a = 3$, $b = 5$ then equation (1) and (2) have no solution $\Rightarrow$ No solution exists. $\Rightarrow$ **(D) is correct.]**

Q.15 (BD)

$$\left(\frac{M^2}{2} + M + I \right) \left(\frac{M^2}{2} - M + I \right) = \left(\frac{M^2}{2} + I \right)^2 - M^2$$

$$= \frac{M^4}{4} + I + M^2 - M^2 \quad (\because M^3 = 0)$$

$$= I$$

$$\Rightarrow \det \left(\frac{M^2}{2} + M + I \right) \cdot \det \left(\frac{M^2}{2} - M + I \right) = I \quad \text{Ans.}$$

Q.16 (AB)

$$A_{2 \times p} \cdot B_{3 \times n} = C_{M \times 2}, \quad C_{12} = \sum_{r=1}^3 a_{1r} b_{r2} = \sum_{r=1}^3 \frac{(1+r)(r-2)}{4}$$

$$= \sum_{r=1}^3 \frac{r^2 - r - 2}{4} = \frac{14 - 6 - 6}{4} = \frac{1}{2}$$

$$P = 3, n = 2, m = 2$$

$$\Rightarrow m + n + p = 7$$

$$B_{3 \times n} \cdot A_{2 \times p} = C_{m \times 2} \quad C_{32} = \sum_{r=1}^2 b_{3r} a_{r2} = \sum_{r=1}^2 \frac{(3-r)(r+2)}{4} = \sum_{r=1}^2 \frac{-r^2 + r + 6}{4} = \frac{5}{2}$$

$$m = 3, n = 2, p = 2$$

$$\Rightarrow m - n + p = 3$$

Q.17 (BCD)

$$A = \begin{bmatrix} x & 1 & 0 \\ 1 & x & 1 \\ 0 & 1 & x \end{bmatrix};$$

$$x(x^2 - 1) - 1(x) = 6 - 2 = 4$$

Q.18 (AD)

$$|A - xI| = 0$$

$$\Rightarrow \begin{vmatrix} 1-x & 2 & 0 \\ 0 & 2-x & 0 \\ 0 & -1 & 2-x \end{vmatrix} = 0$$

$$\Rightarrow (1-x)(2-x)(2-x) = 0$$

$$\Rightarrow x^3 - 5x^2 + 8x - 4 = 0$$

$$\Rightarrow A^3 - 5A^2 + 8A - 4I = 0$$

$$\Rightarrow A^4 - 5A^3 + 8A^2 - 4A = 0$$

$$\Rightarrow A^4 - 5A^3 + 8A^2 - 2A = 2A \quad \dots\dots(i)$$

$$\text{As, } P = A^4 + 8A, \quad Q = -5A^3 + 8A^2 - 10A$$

$$\Rightarrow P + Q = A^4 - 5A^3 + 8A^2 - 2A$$

$$\therefore P + Q = 2A \quad (\text{using Eq.(i)})$$

$$\text{So, } \text{tr}(P + Q) = 2\text{tr } |A| = 10$$

$$\text{and } \det.(P + Q) = |2A| = 8|A| = 32$$

Q.19 [0000]

$$ax^2 + bx + c = 0 \begin{cases} \alpha_1 \\ \alpha_2 \end{cases}$$

$$px^2 + qx + r = 0 \begin{cases} \beta_1 \\ \beta_2 \end{cases}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \quad ; \quad \alpha_1 \alpha_2 = \frac{c}{a}$$

$$\beta_1 + \beta_2 = -\frac{q}{p} ; \beta_1 \beta_2 = \frac{r}{p}$$

for non-trivial solution

$$\begin{vmatrix} \alpha_1 & \alpha_2 \\ \beta_1 & \beta_2 \end{vmatrix} = 0 \Rightarrow \alpha_1 \beta_2 = \beta_1 \alpha_2$$

$$\text{or } \frac{\alpha_1}{\alpha_2} = \frac{\beta_1}{\beta_2}$$

$$\text{or } \frac{\alpha_1 + \alpha_2}{\alpha_1 - \alpha_2} = \frac{\beta_1 + \beta_2}{\beta_1 - \beta_2} \quad (\text{Using componendo and dividendo})$$

$$\therefore \frac{(\alpha_1 + \alpha_2)^2}{(\alpha_1 + \alpha_2)^2 - 4\alpha_1\alpha_2} = \frac{(\beta_1 + \beta_2)^2}{(\beta_1 + \beta_2)^2 - 4\beta_1\beta_2}$$

$$\frac{\frac{b^2}{a^2}}{\frac{b^2}{a^2} - \frac{4c}{a}} = \frac{\frac{q^2}{r^2}}{\frac{q^2}{p^2} - \frac{4r}{p}}$$

$$\frac{b^2}{b^2 - 4ac} = \frac{q^2}{q^2 - 4pr}$$

$$\frac{b^2}{q^2} = \frac{b^2 - 4ac}{q^2 - 4pr} = \frac{ac}{pr}$$

Q.20 [2]

$$x^6 (x^3 - 6x^2 + 12x - 8) = 0 \Rightarrow x = 0, 0, 0, 0, 0, 0, 2, 2, 2$$

$$\therefore |A| = \begin{vmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{vmatrix} = 8$$

$$\text{Now, } b_{11} = b_{22} = b_{33} = 6$$

$$B = \begin{bmatrix} 6 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{bmatrix} \Rightarrow |B| = 6^3$$

$$\text{Now, } \frac{|\text{adj } B|}{|\text{adj}(\text{adj}(\text{adj } A))|} = \frac{|B|^2}{|A|^{2^3}} = \frac{6^6}{8^8} = \frac{6^6}{2^{24}} = \left(\frac{6}{16}\right)^6 = \left(\frac{3}{8}\right)^6$$

$$\therefore q - 2p = 8 - 6 = 2. \text{ Ans.}$$

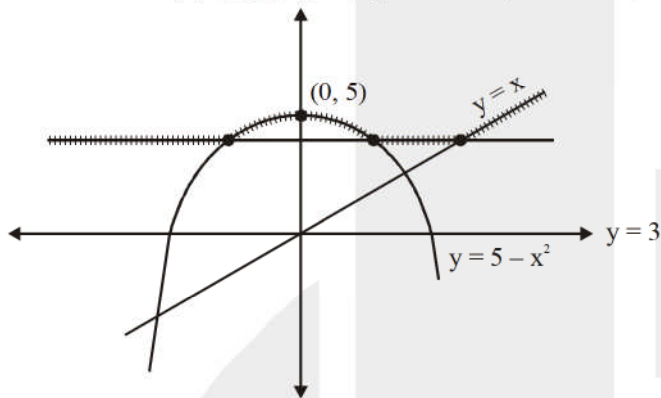
Q.21 [3]

$\because$ A is orthogonal invertible matrix

$$\therefore A^{-1} = A^T$$



$f(x) = \text{tr.}(A^{-1}) = \text{tr.}(A^T)$
 B is involutory invertible matrix $= 5 - x^2$
 $\therefore B^{-1} = B \Rightarrow \text{tr.}(B^{-1}) = \text{tr.}(B)$
 $\therefore g(x) = x$
 C is invertible idempotent matrix
 $\therefore C^{-1} = I$
 $\therefore \text{tr.}(C^{-1}) = 3$
 $\therefore \max.(f(x), g(x), h(x)) = \max.(5 - x^2, x, 3)$



$\therefore$ Number of points of non derivability = 3 **Ans.**

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